

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Bachelor of Technology (Artificial Intelligence (AI) and Data Science) SEMESTER - 6 Winter 2025 ( Supply. )**

**Course :Bachelor of Technology (Artificial Intelligence (AI) and Data Science) Branch : Engineering and Technology**

**Semester : SEMESTER - 6**

**Subject Code & Name: BTAIOE604C - AGILE METHODOLOGY**

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**Time : 3 Hours]**

**[Total Marks : 60**

**Instructions to the Students:**

1. Each question carries 12 marks.
  2. Question No. 1 will be compulsory and include objective-type questions.
  3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
  4. Use of non-programmable scientific calculators is allowed.
  5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12**
- 1 Which theory of Agile management emphasizes adaptability and continuous learning?  
a) Scientific management      b) Bureaucratic management  
c) Complexity theory      d) Classical management
  - 2 Story cards in Agile mainly support:  
a) Knowledge sharing and requirement understanding  
b) Financial estimation      c) Code generation      d) Risk modelling
  - 3 Which challenge is commonly faced when migrating to Agile methodologies?  
a) Excessive automation      b) Cultural resistance to change  
c) Over-documentation      d) Lack of tools
  - 4 In Test-Driven Development, refactoring is performed to:  
a) Add new tests      b) Fix failed tests  
c) Improve code quality while keeping tests green  
d) Remove automation
  - 5 In Feature Driven Development (FDD), financial metrics are MOST useful for:  
a) Tracking feature-based cost and value delivery  
b) Technical debt estimation      c) Developer performance appraisal  
d) Sprint retrospectives
  - 6 Which Agile value promotes direct communication within teams?  
a) Processes over people      b) Documentation over code  
c) Tools over interaction  
d) Individuals and interactions over processes and tools
  - 7 Concurrency in Agile requirements generation means:  
a) Requirements are frozen early  
b) Requirements, design, and development progress together  
c) Requirements are handled only by analysts  
d) Testing starts after deployment

- 8 At higher levels of the Story-Card Maturity Model (SMM), story cards primarily act as:
  - a) Informal reminders
  - b) Documentation substitutes
  - c) Planning constraints
  - d) Shared organizational knowledge assets
- 9 Which role in Extreme Programming is responsible for defining user stories?
  - a) Programmer
  - b) Coach
  - c) Customer
  - d) Tracker
- 10 Agile prioritization mainly focuses on:
  - a) Technical complexity
  - b) Developer preference
  - c) Documentation size
  - d) Business value and customer feedback
- 11 Agile QA in global software development mainly relies on:
  - a) Continuous communication and collaboration tools
  - b) Heavy documentation
  - c) Delayed integration
  - d) Centralized control
- 12 Crystal methodology primarily categorizes projects based on:
  - a) Technology stack
  - b) Team size and system criticality
  - c) Customer location
  - d) Development cost

Q2. Solve the following.

- A) Define Agile Software Development and list its core characteristics. Explain the principles of the Agile Manifesto with suitable examples. 6
- B) List different classifications of Agile methods with examples. Describe any FOUR differences between traditional and Agile development models. 6

Q3. Solve the following.

- A) Define Feature Driven Development (FDD). Demonstrate how FDD can be used for feature-centric development. 6
- B) State key practices of Extreme Programming. Explain in detail the lifecycle of Extreme Programming. 6

Q4. Solve Any Two of the following.

- A) Define Agile Information Systems. Explain in detail the Agile decision-making mechanisms. 6
- B) Describe how institutional knowledge evolves through the stages of development and refinement. 6
- C) A software company follows Agile development with teams working remotely from different locations and time zones. Although the teams use Agile practices and online collaboration tools, they face problems in sharing knowledge effectively. As a result, some work is duplicated, Agile practices are applied inconsistently, and new team members take longer to learn. Apply knowledge distribution techniques in Agile to improve decision-making in a distributed team. 6

Q5. Solve Any Two of the following.

- A) List common Agile requirements elicitation techniques. Explain requirements management in an Agile environment. 6

- B) A product-based software company develops and maintains a web-based application using Agile methodologies. The company releases new features frequently based on changing customer needs and market demands. However, the team faces challenges in managing and updating requirements, prioritizing features, and ensuring that customer feedback is effectively incorporated into development. Design an Agile Requirements Engineering workflow for a product-based company. 6
- C) Define the Agile Requirements Abstraction Model. Explain the variance in Agile Requirements Engineering practices. 6
- Q6. Solve Any Two of the following.
- A) Define Test-Driven Development. Describe the lifecycle of Test-Driven Development with a neat, suitable diagram. 6
- B) Define Agile Quality Assurance. Describe quality assurance challenges in global Agile teams. 6
- C) A software development team is building a small application that allows users to track their daily tasks. The team follows Agile practices and wants to ensure the module is reliable, maintainable, and easy to modify in the future. To achieve this, the team decides to implement Test-Driven Development (TDD). As a developer on this team, explain how you would apply Test-Driven Development to develop a simple task-tracking module. The answer should include the step-by-step TDD process, how to write test cases before coding, the role of refactoring, and how this approach improves code quality and supports Agile principles. 6

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